

COMPREHENSIVE LECTURE NOTES & QUESTION BANK

Unit-4: Probability and Theoretical Distributions

Course Component: Advanced Engineering & Statistical Mathematics

Scope: Foundational Concepts, Laws of Probability, Conditional Frameworks, Bayes' Theorem, and Discrete & Continuous Probability Distributions (Binomial, Poisson, Normal)

Features: Rigorous Proofs, Exhaustive Explanations, and Massive Problem Sets with Detailed Step-by-Step Solutions

Chapter 1: Foundational Concepts & Probability Frameworks

1.1 Introduction to Random Phenomena

In deterministic models, the outcome of an experiment or process is completely determined by its initial parameters and physical laws. However, in real-world engineering, scientific, and business applications, systems are subject to inherent variability, noise, and incomplete data. Probability theory provides the formal mathematical language required to model, evaluate, and interpret these non-deterministic systems, which are called **random experiments**.

A random experiment is any process or action whose outcome cannot be predicted with absolute certainty prior to its execution, yet satisfies the following structural criteria:

- It can be repeated infinitely under ostensibly identical, controlled conditions.
- All possible distinct outcomes can be explicitly listed or conceptualized in advance.
- The specific outcome of any individual trial is unpredictable, but across a massive sequence of repetitions, a stable statistical regularity or frequency emerges.

1.2 Sample Space and Events

The mathematical structure of probability rests upon defining the set of all conceivable configurations of a random experiment.

Definition 1.1 (Sample Space): The sample space, denoted by S or Ω , is the exhaustive set of all possible, mutually exclusive outcomes of a random experiment. Each discrete element within this set is termed a sample point or elementary outcome, denoted by ω .

Sample spaces are categorized based on their cardinality:

1. Discrete Sample Spaces:

- *Finite Sample Spaces:* Contain a countable, bounded number of elements. For instance, tossing a fair coin twice yields $S = \{HH, HT, TH, TT\}$, where $|S| = 4$.
- *Countably Infinite Sample Spaces:* Contain elements that can be placed in a one-to-one correspondence with the set of natural numbers. For instance, tossing a coin repeatedly until the first head appears results in $S = \{1, 2, 3, 4, \dots\}$.

2. **Continuous Sample Spaces:** Contain an uncountably infinite number of elements, typically representing measurements along a continuous interval or multidimensional space. For instance, measuring the exact life span t of an electronic semiconductor component yields $S = \{t \in \mathbb{R} \mid t \geq 0\}$.

Definition 1.2 (Event Space): An event A is a subset of the sample space S (i.e., $A \subseteq S$). We state that the event A occurs if the actual realized outcome ω of the experiment belongs to A .

- **Simple (Elementary) Event:** An event containing exactly one sample point, $A = \{\omega\}$.
- **Compound Event:** An event containing multiple sample points, reflecting a more complex condition.
- **Certain Event:** The entire sample space S itself. Since any outcome must reside in S , $P(S) = 1$.
- **Impossible Event:** The empty set, denoted by $\emptyset$. It contains no outcomes, so $P(\emptyset) = 0$.

1.3 Set Operations and Algebra of Events

Since events are subsets of S , we manipulate them using classical set-theoretic operations. Let A and B be events in S :

- **Union ($A \cup B$):** The event consisting of all sample points that belong to A , to B , or to both. This represents the logical condition "at least one of the events occurs" (A OR B).
- **Intersection ($A \cap B$):** The event consisting of all sample points that belong simultaneously to both A and B . This represents the logical condition "both events occur concurrently" (A AND B). If $A \cap B = \emptyset$, the events are declared **mutually exclusive** or **disjoint**.
- **Complement (A^c or A'):** The event containing all sample points in S that do not belong to A . This represents the logical negation "event A does not occur" (NOT A).
- **Difference ($A \setminus B$ or $A - B$):** The event containing all sample points that belong to A but not to B . Note that $A \setminus B = A \cap B^c$.

De Morgan's Laws:

$$(A \cup B)^c = A^c \cap B^c \quad (A \cap B)^c = A^c \cup B^c$$

1.4 Formal Definitions of Probability

1.4.1 Classical (A Priori) Definition

Formulated primarily by Pierre-Simon Laplace, this framework assumes that the sample space is finite and that all elementary outcomes are **equally likely** due to symmetry or physical configuration.

$$P(A) = |A| / |S| = (\text{Number of outcomes favorable to } A) / (\text{Total number of possible outcomes})$$

Limitations: This classical approach breaks down entirely when the sample space is infinite or continuous, or when outcomes are not equally likely (e.g., a biased coin or asymmetrical die).

1.4.2 Relative Frequency (Empirical) Definition

This definition conceptualizes probability as the long-run limiting behavior of an empirical system. If an experiment is replicated N times under identical conditions, and event A occurs exactly f_A times, the relative frequency of A is f_A / N . The probability is defined as:

$$P(A) = \lim_{(N \rightarrow \infty)} (f_A / N)$$

Limitations: It is physically impossible to perform an infinite number of trials, and the mathematical existence of this limit must be assumed as an empirical property rather than a structural proof.

1.4.3 Axiomatic Definition (Kolmogorov Framework)

Developed by Andrey Kolmogorov in 1933, this rigorous approach treats probability as a purely mathematical measure defined over a σ -algebra of events. Given a sample space S and an associated event space, a probability function P is a real-valued function mapping events to real numbers that strictly satisfies the following three axioms:

- **Axiom 1 (Non-negativity):** For any event A , $P(A) \geq 0$.
- **Axiom 2 (Normalization):** The probability of the certain event S is exactly 1, i.e., $P(S) = 1$.
- **Axiom 3 (Countable Additivity):** If $\{A_1, A_2, A_3, \dots\}$ is a sequence of pairwise mutually exclusive (disjoint) events, such that $A_i \cap A_j = \emptyset$ for all $i \neq j$, then:

$$P(\cup_{(i=1 \text{ to } \infty)} A_i) = \sum_{(i=1 \text{ to } \infty)} P(A_i)$$

Immediate Corollaries Derived from Kolmogorov's Axioms:

1. **Probability of the Empty Set:** $P(\emptyset) = 0$.

Proof: Let $A_1 = S$ and $A_i = \emptyset$ for all $i \geq 2$. These sets are pairwise disjoint. By Axiom 3, $P(\cup A_i) = P(S) + \sum_{(i=2 \text{ to } \infty)} P(\emptyset)$. Since $\cup A_i = S$, we have $P(S) = P(S) + \sum P(\emptyset)$, which implies $\sum P(\emptyset) = 0$. By Axiom 1, probabilities are non-negative, hence $P(\emptyset) = 0$.

2. **Complement Rule:** For any event A , $P(A^C) = 1 - P(A)$.

Proof: By definition, $A \cup A^C = S$ and $A \cap A^C = \emptyset$. Applying Axiom 3 to these two disjoint sets gives $P(A \cup A^C) = P(A) + P(A^C)$. Substituting $P(S) = 1$ from Axiom 2 yields $1 = P(A) + P(A^C)$, or $P(A^C) = 1 - P(A)$.

3. **Bounded Monotonicity:** For any event A , $0 \leq P(A) \leq 1$.

4. **Subsets / Monotonicity Property:** If $A \subseteq B$, then $P(A) \leq P(B)$.

Proof: We can decompose B into two disjoint sets: $B = A \cup (B \setminus A)$, where $B \setminus A = B \cap A^C$. Since A and $(B \setminus A)$ are disjoint, Axiom 3 gives $P(B) = P(A) + P(B \setminus A)$. Since Axiom 1 dictates $P(B \setminus A) \geq 0$, it follows directly that $P(B) \geq P(A)$.

1.5 Extensive Problem Set: Foundational Theory & Sample Spaces

Problem 1.1

An electronic manufacturing line tests three newly completed microprocessors sequentially. A processor is classified as "D" if it is defective, and "G" if it is good. 1. Explicitly outline the full sample space S . 2. Define event A as "at least two processors are defective". 3. Define event B as "the first processor inspected is good". 4. Find the components and probabilities of A , B , $A \cup B$, and $A \cap B$ assuming all simple outcomes are equally likely.

Solution:

1. Each processor inspection has 2 outcomes. For 3 processors, the sample space contains $2^3 = 8$ outcomes:

$$S = \{GGG, GGD, GDG, GDD, DGG, DGD, DDG, DDD\}$$

2. Event A requires 2 or 3 defectives (D):

$$A = \{GDD, DGD, DDG, DDD\} \Rightarrow |A| = 4$$

3. Event B requires the first letter to be G:

$$B = \{GGG, GGD, GDG, GDD\} \Rightarrow |B| = 4$$

4. Calculating the union and intersection:

- $A \cup B = \{GDD, DGD, DDG, DDD, GGG, GGD, GDG\}$ (contains 7 outcomes).
- $A \cap B = \{GDD\}$ (contains 1 outcome).

Since outcomes are equally likely, each outcome has probability $1/8$:

$$P(A) = 4/8 = 0.50$$

$$P(B) = 4/8 = 0.50$$

$$P(A \cup B) = 7/8 = 0.875$$

$$P(A \cap B) = 1/8 = 0.125$$

Problem 1.2

An automated warehouse shipping system routes packages to three bays. The probability that a package is misrouted on a given day is $P(E_1) = 0.08$ for Bay 1, $P(E_2) = 0.05$ for Bay 2, and $P(E_3) = 0.04$ for Bay 3. Assume the events are mutually exclusive. Calculate the probability that a package processed today is routed correctly through all three bays.

Solution:

Let M be the event that a package is misrouted. Since routing failures across the bays are mutually exclusive, the total probability of a misrouting error anywhere in the system is the sum of the individual probabilities:

$$P(M) = P(E_1 \cup E_2 \cup E_3) = P(E_1) + P(E_2) + P(E_3) = 0.08 + 0.05 + 0.04 = 0.17$$

The event that a package is routed correctly is the complement of a misrouting error (M^c):

$$P(M^c) = 1 - P(M) = 1 - 0.17 = 0.83$$

Problem 1.3

Prove analytically from Kolmogorov axioms that for any two arbitrary events X and Y , $P(X \cap Y^c) = P(X) - P(X \cap Y)$.

Solution:

We can decompose event X into two disjoint sets: those points that belong to both X and Y , and those points that belong to X but not to Y .

$$X = (X \cap Y) \cup (X \cap Y^c)$$

To verify they are disjoint, we check their intersection:

$$(X \cap Y) \cap (X \cap Y^c) = X \cap (Y \cap Y^c) = X \cap \emptyset = \emptyset$$

Since they are disjoint, we apply Axiom 3 (Finite Additivity derived from Countable Additivity):

$$P(X) = P(X \cap Y) + P(X \cap Y^c)$$

Rearranging the terms directly yields the desired identity:

$$P(X \cap Y^c) = P(X) - P(X \cap Y)$$

Problem 1.4

A continuous sample space is defined by the interval $S = \{x \in \mathbb{R} \mid 0 \leq x \leq 10\}$. The probability of an event falling within any sub-interval $[a, b]$ is proportional to the length of that sub-interval, scaled by $1/10$. Let event $A = [1, 4]$ and event $B = [3, 7]$. Calculate $P(A^c \cap B^c)$.

Solution:

By De Morgan's Law, we know that $A^c \cap B^c = (A \cup B)^c$. Let's first determine the union $A \cup B$:

$$A \cup B = [1, 4] \cup [3, 7] = [1, 7]$$

The total length of this interval is $7 - 1 = 6$. Therefore, its probability is:

$$P(A \cup B) = 6 / 10 = 0.60$$

Now, using the complement rule:

$$P(A^c \cap B^c) = P((A \cup B)^c) = 1 - P(A \cup B) = 1 - 0.60 = 0.40$$

Problem 1.5

In a quality control analysis of 200 integrated circuits, 30 have surface finish blemishes, 25 have propagation delay defects, and 10 have both types of flaws. If an integrated circuit is selected at random, find the probability that it has at least one flaw.

Solution:

Let F be the event of a surface finish blemish, and D be the event of a propagation delay defect. Using the classical definition of probability, we have:

$$P(F) = 30 / 200 = 0.15$$

$$P(D) = 25 / 200 = 0.125$$

$$P(F \cap D) = 10 / 200 = 0.05$$

The event "at least one flaw" is represented by the union $F \cup D$. Applying the fundamental addition rules:

$$P(F \cup D) = P(F) + P(D) - P(F \cap D) = 0.15 + 0.125 - 0.05 = 0.225$$

Chapter 2: The Laws of Probability

2.1 The Addition Law of Probability

The addition rules allow us to compute the probability of the union of multiple events. We begin with two arbitrary events and expand to more general forms.

Theorem 2.1 (Addition Law for Two General Events):

For any two arbitrary events A and B belonging to a sample space S :

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Proof: We can express the union of two events $A \cup B$ as the union of two mutually exclusive events: A and $(B \setminus A)$. Thus, $A \cup B = A \cup (B \cap A^c)$.

Since A and $(B \cap A^c)$ are disjoint, by Axiom 3:

$$P(A \cup B) = P(A) + P(B \cap A^c) \quad \text{— (Equation 2.1)}$$

From Problem 1.3, we proved that $P(B \cap A^c) = P(B) - P(A \cap B)$. Substituting this back into Equation 2.1 gives:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

This concludes the proof. The subtraction of $P(A \cap B)$ corrects for double counting, as those outcomes are counted twice when summing $P(A)$ and $P(B)$.

Theorem 2.2 (Addition Law for Three General Events):

For any three events A , B , and C belonging to S :

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C)$$

Proof: Group $B \cup C$ as a single compound event, say $E = B \cup C$. Then $A \cup B \cup C = A \cup E$. Applying the two-event addition law:

$$P(A \cup E) = P(A) + P(E) - P(A \cap E)$$

Substitute $E = B \cup C$ back into the equation:

$$P(A \cup B \cup C) = P(A) + P(B \cup C) - P(A \cap (B \cup C))$$

Using the distributive law for set intersection over union: $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$. Now, apply the two-event addition law to both $P(B \cup C)$ and $P((A \cap B) \cup (A \cap C))$:

$$P(B \cup C) = P(B) + P(C) - P(B \cap C)$$

$$P((A \cap B) \cup (A \cap C)) = P(A \cap B) + P(A \cap C) - P((A \cap B) \cap (A \cap C))$$

Since $(A \cap B) \cap (A \cap C) = A \cap B \cap C$, combining these expressions yields:

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(B \cap C) - [P(A \cap B) + P(A \cap C) - P(A \cap B \cap C)]$$

Expanding the brackets gives the final theorem expression.

This principle expands to n events and is formally known as the **Inclusion-Exclusion Principle**.

2.2 Conditional Probability

Conditional probability updates the probability of an event based on new information or an intermediate event occurring. This effectively scales down the active sample space from S to the conditioning event.

Definition 2.1 (Conditional Probability): Let A and B be two events in S such that $P(B) > 0$. The conditional probability of event A given that event B has occurred, denoted $P(A | B)$, is defined as:

$$P(A | B) = P(A \cap B) / P(B)$$

Similarly, if $P(A) > 0$, then $P(B | A) = P(A \cap B) / P(A)$.

2.3 The Multiplication Law of Probability

Rearranging the definition of conditional probability yields the multiplication law, which is used to calculate the probability of multiple events occurring concurrently.

Multiplication Law for Two Events:

$$P(A \cap B) = P(B) \cdot P(A | B) \quad (\text{if } P(B) > 0)$$

$$P(A \cap B) = P(A) \cdot P(B | A) \quad (\text{if } P(A) > 0)$$

Theorem 2.3 (General Chain Rule / Multiplication Law for n Events):

If $A_1, A_2, \dots, A_n$ are events such that $P(A_1 \cap A_2 \cap \dots \cap A_{n-1}) > 0$, then:

$$P(A_1 \cap A_2 \cap A_3 \cap \dots \cap A_n) = P(A_1) \cdot P(A_2 | A_1) \cdot P(A_3 | A_1 \cap A_2) \cdots P(A_n | A_1 \cap A_2 \cap \dots \cap A_{n-1})$$

2.4 Statistical Independence

Definition 2.2 (Independence): Two events A and B are statistically independent if and only if the occurrence of one does not alter the probability of the other. Mathematically, this is true if:

$$P(A | B) = P(A) \quad \text{and} \quad P(B | A) = P(B)$$

Substituting this into the multiplication law gives the structural definition of independence:

$$P(A \cap B) = P(A) \cdot P(B)$$

Mutually Exclusive vs. Independent Events

It is important not to confuse mutually exclusive events with independent events:

- **Mutually Exclusive:** $A \cap B = \emptyset$. The events cannot happen at the same time. If $P(A) > 0$ and $P(B) > 0$, then $P(A \cap B) = 0 \neq P(A) \cdot P(B)$. Therefore, non-trivial mutually exclusive events are always **dependent** because knowing one occurs tells you the other definitely did not.
- **Independent:** $P(A \cap B) = P(A) \cdot P(B)$. The occurrence of one event provides no information about the other.

2.5 Extensive Problem Set: Addition & Multiplication Laws

Problem 2.1

A data communications system sends packets over two redundant lines. The probability that Line 1 experiences a failure today is **0.12**, the probability that Line 2 fails is **0.15**, and the probability that both lines fail concurrently is **0.04**. Find the probability that: 1. At least one line fails during the day. 2. Only Line 1 fails.

Solution:

Let L_1 and L_2 be the events that Line 1 and Line 2 fail, respectively. We are given: $P(L_1) = 0.12$, $P(L_2) = 0.15$, and $P(L_1 \cap L_2) = 0.04$.

1. The event that "at least one line fails" is the union $L_1 \cup L_2$. Applying the addition law:

$$P(L_1 \cup L_2) = P(L_1) + P(L_2) - P(L_1 \cap L_2) = 0.12 + 0.15 - 0.04 = 0.23$$

2. The event that "only Line 1 fails" means Line 1 fails and Line 2 does not, which is $L_1 \cap L_2^c$. Using the identity from Problem 1.3:

$$P(L_1 \cap L_2^c) = P(L_1) - P(L_1 \cap L_2) = 0.12 - 0.04 = 0.08$$

Problem 2.2

An executive board consists of 6 micro-computing specialists, 4 cybersecurity specialists, and 5 data scientists. A subcommittee of 3 members is chosen at random without replacement. Find the probability that the committee includes exactly one representative from each specialty.

Solution:

The total number of board members is $6 + 4 + 5 = 15$. We select a committee of 3 members. Let's use combinations to find the total number of ways to choose 3 people out of 15:

$$\text{Total Combinations} = {}_{15}C_3 = (15 \cdot 14 \cdot 13) / (3 \cdot 2 \cdot 1) = 455$$

To have exactly one member from each specialty, we select 1 from the 6 computing specialists, 1 from the 4 cybersecurity specialists, and 1 from the 5 data scientists:

$$\text{Favorable Combinations} = {}_6C_1 \cdot {}_4C_1 \cdot {}_5C_1 = 6 \cdot 4 \cdot 5 = 120$$

Using the classical definition of probability, we divide the favorable combinations by the total combinations:

$$P(\text{Exactly One of Each}) = 120 / 455 = 24 / 91 \approx 0.2637$$

Problem 2.3

A pipeline monitoring system uses three independent, sequential sensors. The system flags an alert if Sensor A, Sensor B, or Sensor C detects a pressure drop. The independent detection probabilities are $P(A) = 0.90$, $P(B) = 0.85$, and $P(C) = 0.80$. If a pressure drop occurs, what is the probability that the system flags an alert?

Solution:

The system flags an alert if at least one sensor detects the drop, which is represented by the union $A \cup B \cup C$. Since the sensors operate independently, their complements (failing to detect the drop) are also independent:

$$P(A^c) = 1 - 0.90 = 0.10$$

$$P(B^c) = 1 - 0.85 = 0.15$$

$$P(C^c) = 1 - 0.80 = 0.20$$

The probability that all three sensors fail to detect the drop is the product of their individual failure probabilities:

$$P(A^c \cap B^c \cap C^c) = P(A^c) \cdot P(B^c) \cdot P(C^c) = 0.10 \cdot 0.15 \cdot 0.20 = 0.003$$

The probability that at least one sensor succeeds is the complement of all three failing:

$$P(A \cup B \cup C) = 1 - P(A^c \cap B^c \cap C^c) = 1 - 0.003 = 0.997$$

Problem 2.4

A box contains 8 red balls and 5 white balls. Two balls are drawn sequentially without replacement. Find the probability that the first ball drawn is red and the second ball drawn is white.

Solution:

Let R_1 be the event that the first ball is red, and W_2 be the event that the second ball is white. We want to find the intersection $P(R_1 \cap W_2)$.

On the first draw, there are $8 + 5 = 13$ total balls, 8 of which are red:

$$P(R_1) = 8 / 13$$

Since the drawing is done without replacement, there are 12 balls remaining for the second draw. If a red ball was chosen first, all 5 white balls are still in the box. This gives the conditional probability:

$$P(W_2 | R_1) = 5 / 12$$

Applying the multiplication law:

$$P(R_1 \cap W_2) = P(R_1) \cdot P(W_2 | R_1) = (8 / 13) \cdot (5 / 12) = 40 / 156 = 10 / 39 \approx 0.2564$$

Problem 2.5

The probability that an automated drone completes its delivery flight successfully is **0.95** if the weather is clear, and **0.60** if it is raining. The local weather forecast estimates a **20%** chance of rain today. Find the probability that a drone selected at random completes its delivery successfully today.

Solution:

Let **S** be the event of a successful flight, **R** be the event of rain, and **C** be the event of clear weather (R^C). We are given the following probabilities:

$$P(R) = 0.20 \Rightarrow P(C) = 1 - 0.20 = 0.80$$

$$P(S | C) = 0.95$$

$$P(S | R) = 0.60$$

A successful flight can occur under either clear weather or rainy weather. Since these weather conditions are mutually exclusive, we can express **S** as the sum of two disjoint intersections:

$$P(S) = P(S \cap C) + P(S \cap R)$$

Applying the multiplication law to both terms:

$$P(S) = P(C) \cdot P(S | C) + P(R) \cdot P(S | R)$$

$$P(S) = (0.80 \cdot 0.95) + (0.20 \cdot 0.60) = 0.76 + 0.12 = 0.88$$

Chapter 3: Law of Total Probability & Bayes' Theorem

3.1 The Partition of a Sample Space

To analyze complex conditional systems, it helps to divide the sample space into distinct, non-overlapping pieces.

Definition 3.1 (Partition): A collection of events $\{B_1, B_2, \dots, B_n\}$ forms a partition of the sample space S if the following two conditions are met:

1. The events are pairwise mutually exclusive: $B_i \cap B_j = \emptyset$ for all $i \neq j$.
2. The collection is collectively exhaustive, meaning their union restores the entire sample space:

$$B_1 \cup B_2 \cup \dots \cup B_n = \bigcup_{(i=1 \text{ to } n)} B_i = S$$

3.2 The Law of Total Probability

Theorem 3.1 (Law of Total Probability):

Let $\{B_1, B_2, \dots, B_n\}$ be a partition of S such that $P(B_i) > 0$ for all i . For any arbitrary event A in S :

$$P(A) = \sum_{(i=1 \text{ to } n)} P(B_i) \cdot P(A | B_i)$$

Proof: Since $\{B_1, B_2, \dots, B_n\}$ partitions S , we can write $A = A \cap S$. Substituting the partition union for S gives:

$$A = A \cap (B_1 \cup B_2 \cup \dots \cup B_n)$$

Using the distributive law of set theory:

$$A = (A \cap B_1) \cup (A \cap B_2) \cup \dots \cup (A \cap B_n)$$

Because the events B_i are mutually exclusive, the intersections $(A \cap B_i)$ are also mutually exclusive. Applying the finite additivity property from Axiom 3:

$$P(A) = \sum_{(i=1 \text{ to } n)} P(A \cap B_i)$$

Finally, using the multiplication law to substitute $P(A \cap B_i) = P(B_i) \cdot P(A | B_i)$ gives the final result:

$$P(A) = \sum_{(i=1 \text{ to } n)} P(B_i) \cdot P(A | B_i)$$

3.3 Bayes' Theorem

Bayes' Theorem provides a way to invert conditional probabilities. It calculates the probability that a specific partition component B_k caused event A to occur. This is often referred to as updating a **prior probability** $P(B_k)$ to a **posterior probability** $P(B_k | A)$ after observing evidence A .

Theorem 3.2 (Bayes' Theorem):

Let $\{B_1, B_2, \dots, B_n\}$ be a partition of S , and let A be an event with $P(A) > 0$. For any specific index k ($1 \leq k \leq n$):

$$P(B_k | A) = [P(B_k) \cdot P(A | B_k)] / [\sum_{i=1 \text{ to } n} P(B_i) \cdot P(A | B_i)]$$

Proof: By the definition of conditional probability:

$$P(B_k | A) = P(B_k \cap A) / P(A)$$

Using the multiplication law in the numerator gives $P(B_k \cap A) = P(B_k) \cdot P(A | B_k)$. Substituting the Law of Total Probability for the denominator $P(A)$ yields Bayes' Theorem.

3.4 Extensive Problem Set: Total Probability & Bayes' Frameworks

Problem 3.1

A manufacturing plant uses three different machinery configurations (M_1 , M_2 , M_3) to produce structural rivets. M_1 handles 50% of total production, M_2 handles 30%, and M_3 handles 20%. The defective rates for each machine are 1%, 2%, and 4%, respectively. 1. Find the probability that a randomly selected rivet is defective. 2. If a chosen rivet is defective, what is the probability it was produced by machine M_3 ?

Solution:

Let D be the event that a rivet is defective. The production shares give our prior probabilities for each machine:

$$P(M_1) = 0.50, \quad P(M_2) = 0.30, \quad P(M_3) = 0.20$$

The conditional defect rates are:

$$P(D | M_1) = 0.01, \quad P(D | M_2) = 0.02, \quad P(D | M_3) = 0.04$$

Step 1: Calculate the total probability of a defect, $P(D)$:

$$P(D) = P(M_1)P(D | M_1) + P(M_2)P(D | M_2) + P(M_3)P(D | M_3)$$

$$P(D) = (0.50 \cdot 0.01) + (0.30 \cdot 0.02) + (0.20 \cdot 0.04) = 0.005 + 0.006 + 0.008 = 0.019$$

The overall defect rate for the plant is 1.9%.

Step 2: Apply Bayes' Theorem to find $P(M_3 | D)$:

$$P(M_3 | D) = [P(M_3) \cdot P(D | M_3)] / P(D) = (0.20 \cdot 0.04) / 0.019 = 0.008 / 0.019 = 8 / 19 \approx 0.4211$$

There is a 42.11% chance that the defective rivet came from machine M_3 .

Problem 3.2

A medical screening test for a rare disease has a sensitivity of **99%** (probability of testing positive given the patient has the disease) and a specificity of **95%** (probability of testing negative given the patient does not have the disease). The disease is present in **0.5%** of the population. If a patient tests positive, what is the probability that they actually have the disease?

Solution:

Let **Inf** be the event that a patient is infected, and **Well** be the event that they are healthy (Inf^c). Let **Pos** be the event of a positive test result. We are given:

$$P(\text{Inf}) = 0.005 \Rightarrow P(\text{Well}) = 1 - 0.005 = 0.995$$

$$P(\text{Pos} \mid \text{Inf}) = 0.99 \text{ (Sensitivity)}$$

$$P(\text{Pos}^c \mid \text{Well}) = 0.95 \Rightarrow P(\text{Pos} \mid \text{Well}) = 1 - 0.95 = 0.05 \text{ (False Positive Rate)}$$

First, calculate the total probability of a positive test result, $P(\text{Pos})$:

$$P(\text{Pos}) = P(\text{Inf})P(\text{Pos} \mid \text{Inf}) + P(\text{Well})P(\text{Pos} \mid \text{Well})$$

$$P(\text{Pos}) = (0.005 \cdot 0.99) + (0.995 \cdot 0.05) = 0.00495 + 0.04975 = 0.05470$$

Now, use Bayes' Theorem to find the probability of infection given a positive result, $P(\text{Inf} \mid \text{Pos})$:

$$P(\text{Inf} \mid \text{Pos}) = [P(\text{Inf}) \cdot P(\text{Pos} \mid \text{Inf})] / P(\text{Pos}) = 0.00495 / 0.05470 = 495 / 5470 \approx 0.0905$$

Even with a highly accurate test, the probability of actually having the disease given a positive result is only **9.05%**. This is because the disease is very rare in the general population.

Problem 3.3

In a financial software company, three teams write code: Team Alpha writes **40%**, Team Beta writes **35%**, and Team Gamma writes **25%**. The probabilities of introducing a critical security vulnerability during development are **0.05**, **0.03**, and **0.02** for each team, respectively. A security audit identifies a critical vulnerability in a section of code. Determine which team is most likely responsible.

Solution:

Let V be the event that a code module contains a vulnerability. The team prior probabilities and conditional risk profiles are:

$$P(\text{Alpha}) = 0.40, \quad P(\text{Beta}) = 0.35, \quad P(\text{Gamma}) = 0.25$$

$$P(V | \text{Alpha}) = 0.05, \quad P(V | \text{Beta}) = 0.03, \quad P(V | \text{Gamma}) = 0.02$$

First, calculate the total probability of a vulnerability, $P(V)$:

$$P(V) = (0.40 \cdot 0.05) + (0.35 \cdot 0.03) + (0.25 \cdot 0.02) = 0.020 + 0.0105 + 0.005 = 0.0355$$

Now, use Bayes' Theorem to compute the posterior probabilities for each team:

$$P(\text{Alpha} | V) = 0.0200 / 0.0355 = 200 / 355 \approx 0.5634$$

$$P(\text{Beta} | V) = 0.0105 / 0.0355 = 105 / 355 \approx 0.2958$$

$$P(\text{Gamma} | V) = 0.0050 / 0.0355 = 50 / 355 \approx 0.1408$$

Comparing the results shows that **Team Alpha** is the most likely responsible party, with a posterior probability of **56.34%**.

Problem 3.4

An asset management company classifies stock performance into three categories: Upper Trend (U), Stable (S), and Lower Trend (L). The historical distribution of stocks in their portfolio is $P(U) = 0.25$, $P(S) = 0.60$, and $P(L) = 0.15$. An automated trading algorithm issues a "BUY" signal based on technical indicators. The conditional probabilities of receiving a BUY signal are $P(BUY | U) = 0.80$, $P(BUY | S) = 0.30$, and $P(BUY | L) = 0.10$. If the algorithm issues a BUY signal for a stock, what is the probability that the stock is actually in an Upper Trend?

Solution:

We want to find $P(U | BUY)$. First, compute the total probability of a BUY signal, $P(BUY)$:

$$P(BUY) = P(U)P(BUY | U) + P(S)P(BUY | S) + P(L)P(BUY | L)$$

$$P(BUY) = (0.25 \cdot 0.80) + (0.60 \cdot 0.30) + (0.15 \cdot 0.10) = 0.200 + 0.180 + 0.015 = 0.395$$

Now, use Bayes' Theorem to find the updated probability for the Upper Trend category:

$$P(U | BUY) = [P(U) \cdot P(BUY | U)] / P(BUY) = 0.200 / 0.395 = 200 / 395 = 40 / 79 \approx 0.5063$$

Given a BUY signal, there is a **50.63%** probability that the stock is in an Upper Trend.

Problem 3.5

A credit risk evaluation system uses an AI model to flag high-risk transactions. Historically, 2% of all transactions are fraudulent. The AI model correctly flags fraudulent transactions 92% of the time. However, it also incorrectly flags legitimate transactions as fraudulent 3% of the time. If the system flags a transaction as fraudulent, find the probability that it is actually legitimate.

Solution:

Let F be the event that a transaction is fraudulent, and L be the event that it is legitimate (F^c). Let $Flag$ be the event that the transaction is flagged. We are given:

$$P(F) = 0.02 \Rightarrow P(L) = 0.98$$

$$P(Flag | F) = 0.92$$

$$P(Flag | L) = 0.03$$

First, calculate the total probability of a transaction being flagged, $P(Flag)$:

$$P(Flag) = P(F)P(Flag | F) + P(L)P(Flag | L)$$

$$P(Flag) = (0.02 \cdot 0.92) + (0.98 \cdot 0.03) = 0.0184 + 0.0294 = 0.0478$$

We want to find the probability that a flagged transaction is actually legitimate, $P(L | Flag)$:

$$P(L | Flag) = [P(L) \cdot P(Flag | L)] / P(Flag) = 0.0294 / 0.0478 = 294 / 478 = 147 / 239 \approx 0.6151$$

There is a **61.51%** probability that a flagged transaction is actually legitimate.

Chapter 4: Discrete Probability Distributions – Binomial Framework

4.1 Mathematical Foundations of Random Variables

To bridge structural set theory and quantitative data analysis, we map sample points to numbers using random variables.

Definition 4.1 (Random Variable): A random variable X is a function that maps each element ω in a sample space S to a real number x (i.e., $X: S \rightarrow \mathfrak{R}$).

For a discrete random variable, the mapping produces a countable set of distinct values. Its behavior is defined by its **Probability Mass Function (PMF)**, denoted $p(x) = P(X = x)$, which satisfies:

1. $p(x_i) \geq 0$ for all possible values x_i .
2. $\sum p(x_i) = 1$.

4.2 Bernoulli Trials and the Binomial Distribution Framework

A **Bernoulli trial** is a random experiment with exactly two possible outcomes: "Success" (S) with probability p , and "Failure" (F) with probability $q = 1 - p$. A Binomial distribution extends this to a series of independent Bernoulli trials.

Definition 4.2 (Binomial Distribution): Let X be a discrete random variable representing the total number of successes in n independent Bernoulli trials, where each trial has a constant success probability p . We say X follows a Binomial distribution, written $X \sim \text{Bin}(n, p)$. Its probability mass function is given by:

$$P(X = x) = {}_n C_x \cdot p^x \cdot q^{n-x} \quad \text{for } x = 0, 1, 2, \dots, n$$

where ${}_n C_x = n! / [x!(n - x)!]$ is the binomial coefficient, representing the number of ways to arrange x successes across n trials.

4.3 Step-by-Step Mathematical Proof of Moments

4.3.1 Expected Value (Mean) Derivation

The expected value of a discrete distribution is given by $E[X] = \sum x \cdot P(X = x)$. For a Binomial distribution:

$$E[X] = \sum_{(x=0 \text{ to } n)} x \cdot \{ n! / [x!(n - x)!] \} \cdot p^x \cdot q^{n-x}$$

The term for $x = 0$ is 0, so we can start the summation at $x = 1$. Since $x / x! = 1 / (x - 1)!$, we can rewrite the equation as:

$$E[X] = \sum_{(x=1 \text{ to } n)} \{ n! / [(x - 1)!(n - x)!] \} \cdot p^x \cdot q^{n-x}$$

Factor out n and p from the summation expression:

$$E[X] = n \cdot p \cdot \sum_{(x=1 \text{ to } n)} \{ (n - 1)! / [(x - 1)!(n - x)!] \} \cdot p^{x-1} \cdot q^{n-x}$$

Let $y = x - 1$ and $m = n - 1$. As x ranges from 1 to n , y ranges from 0 to m . Note that $n - x = (m + 1) - (y + 1) = m - y$. Substituting these variables gives:

$$E[X] = n \cdot p \cdot \sum_{(y=0 \text{ to } m)} \{ m! / [y!(m - y)!] \} \cdot p^y \cdot q^{m-y}$$

The summation is now the expansion of $(p + q)^m$ using the Binomial Theorem. Since $p + q = 1$, then $(p + q)^m = 1^m = 1$. This leaves us with:

$$\text{Mean: } \mu = E[X] = n \cdot p$$

4.3.2 Variance Derivation

To calculate the variance, we use the identity $\text{Var}(X) = E[X^2] - (E[X])^2$. First, let's find $E[X(X-1)] = E[X^2] - E[X]$:

$$E[X(X-1)] = \sum_{(x=0 \text{ to } n)} x(x - 1) \cdot \{ n! / [x!(n - x)!] \} \cdot p^x \cdot q^{n-x}$$

The terms for $x = 0$ and $x = 1$ are 0, so we can start the summation at $x = 2$. Since $x(x - 1) / x! = 1 / (x - 2)!$, we get:

$$E[X(X-1)] = \sum_{(x=2 \text{ to } n)} \{ n! / [(x - 2)!(n - x)!] \} \cdot p^x \cdot q^{n-x}$$

Factor out $n(n - 1)$ and p^2 :

$$E[X(X-1)] = n(n - 1)p^2 \cdot \sum_{(x=2 \text{ to } n)} \{ (n - 1)! / [(x - 2)!(n - x)!] \} \cdot p^{x-2} \cdot q^{n-x}$$

Let $y = x - 2$ and $m = n - 2$. This transforms the summation into the binomial expansion for $(p + q)^m$, which equals 1:

$$E[X(X-1)] = n(n - 1)p^2(1) = n^2p^2 - np^2$$

Now, solve for $E[X^2]$:

$$E[X^2] = E[X(X-1)] + E[X] = n^2p^2 - np^2 + np$$

Substitute this back into the variance formula:

$$\text{Var}(X) = (n^2p^2 - np^2 + np) - (np)^2 = np - np^2 = np(1 - p)$$

$$\text{Variance: } \sigma^2 = n \cdot p \cdot q$$

4.4 Extensive Problem Set: Binomial Distribution Problems

Problem 4.1

An production line produces components with a constant historical defect probability of **0.05**. A random sample of 12 components is selected for testing. Calculate the probability that the sample contains: 1. Exactly 2 defective components. 2. At most 1 defective component. 3. More than 2 defective components.

Solution:

Let X be the number of defective components. The trials are independent, so $X \sim \text{Bin}(n = 12, p = 0.05)$. Here, $q = 1 - 0.05 = 0.95$.

1. Probability of exactly 2 defectives ($P(X = 2)$):

$$P(X = 2) = {}_{12}C_2 \cdot (0.05)^2 \cdot (0.95)^{10}$$

$${}_{12}C_2 = (12 \cdot 11) / 2 = 66$$

$$(0.05)^2 = 0.0025, \quad (0.95)^{10} \approx 0.598737$$

$$P(X = 2) = 66 \cdot 0.0025 \cdot 0.598737 \approx 0.0988$$

2. Probability of at most 1 defective ($P(X \leq 1)$):

$$P(X \leq 1) = P(X = 0) + P(X = 1)$$

$$P(X = 0) = {}_{12}C_0 (0.05)^0 (0.95)^{12} = 1 \cdot 1 \cdot 0.540360 = 0.5404$$

$$P(X = 1) = {}_{12}C_1 (0.05)^1 (0.95)^{11} = 12 \cdot 0.05 \cdot 0.568800 = 0.3413$$

$$P(X \leq 1) = 0.5404 + 0.3413 = 0.8817$$

3. Probability of more than 2 defectives ($P(X > 2)$):

Using the complement rule:

$$P(X > 2) = 1 - P(X \leq 2) = 1 - [P(X \leq 1) + P(X = 2)]$$

$$P(X > 2) = 1 - [0.8817 + 0.0988] = 1 - 0.9805 = 0.0195$$

Problem 4.2

A corporate network engine determines that **70%** of data transfers complete within their standard latency window. If 8 data transfers are initiated independently, find the probability that at least 6 transfers complete within this standard window.

Solution:

Let X be the number of successful transfers. Here, $X \sim \text{Bin}(n = 8, p = 0.70)$ and $q = 0.30$. We want to find $P(X \geq 6)$:

$$P(X \geq 6) = P(X = 6) + P(X = 7) + P(X = 8)$$

Let's calculate each term:

$$P(X = 6) = {}_8C_6 \cdot (0.7)^6 \cdot (0.3)^2 = 28 \cdot 0.117649 \cdot 0.09 = 0.2965$$

$$P(X = 7) = {}_8C_7 \cdot (0.7)^7 \cdot (0.3)^1 = 8 \cdot 0.082354 \cdot 0.3 = 0.1977$$

$$P(X = 8) = {}_8C_8 \cdot (0.7)^8 \cdot (0.3)^0 = 1 \cdot 0.057648 \cdot 1 = 0.0576$$

Summing these probabilities:

$$P(X \geq 6) = 0.2965 + 0.1977 + 0.0576 = 0.5518$$

There is a **55.18%** probability that at least 6 transfers complete within the standard window.

Problem 4.3

A safety radar tracking system consists of multiple independent, redundant units. The probability that an individual unit fails to track an incoming object is **0.10**. Find the minimum number of redundant units required to guarantee a **99.9%** probability that at least one unit successfully tracks the object.

Solution:

Let n be the number of units, and X be the number of units that successfully track the object. The success probability for an individual unit is $p = 1 - 0.10 = 0.90$, so $X \sim \text{Bin}(n, 0.90)$. We want to find n such that $P(X \geq 1) \geq 0.999$.

Using the complement rule, we can rewrite $P(X \geq 1)$ as:

$$P(X \geq 1) = 1 - P(X = 0) = 1 - (0.10)^n$$

Set up the inequality:

$$1 - (0.10)^n \geq 0.999 \Rightarrow (0.10)^n \leq 0.001$$

Since $0.001 = (0.10)^3$, we can solve for n directly:

$$n \geq 3$$

A minimum of **3 redundant units** is required.

Problem 4.4

A standardized multiple-choice exam has 20 questions, and each question has 4 options. A student guesses the answers completely at random. Find the mean, variance, and standard deviation of the number of correct answers.

Solution:

Let X be the number of correct answers. Since the student guesses randomly, the success probability for each question is $p = 1/4 = 0.25$. The total number of questions is $n = 20$, so $X \sim \text{Bin}(n = 20, p = 0.25)$ and $q = 0.75$.

Applying the moment formulas derived in section 4.3:

$$\text{Mean } (\mu) = n \cdot p = 20 \cdot 0.25 = 5$$

$$\text{Variance } (\sigma^2) = n \cdot p \cdot q = 20 \cdot 0.25 \cdot 0.75 = 3.75$$

$$\text{Standard Deviation } (\sigma) = \sqrt{3.75} \approx 1.9365$$

Problem 4.5

A manufacturing system experiences minor tool wear failures during operation. The system is configured to run batches of 15 cycles. If the probability of a tool wear failure in a given cycle is **0.20**, calculate the probability that a batch contains exactly 3 failures.

Solution:

Let X be the number of failures per batch, where $X \sim \text{Bin}(n = 15, p = 0.20)$ and $q = 0.80$. We evaluate the PMF at $x = 3$:

$$P(X = 3) = {}_{15}C_3 \cdot (0.20)^3 \cdot (0.80)^{12}$$

$${}_{15}C_3 = (15 \cdot 14 \cdot 13) / (3 \cdot 2 \cdot 1) = 455$$

$$(0.20)^3 = 0.008, \quad (0.80)^{12} \approx 0.068719$$

$$P(X = 3) = 455 \cdot 0.008 \cdot 0.068719 \approx 0.2501$$

There is a **25.01%** probability of observing exactly 3 failures in a batch.

Chapter 5: Discrete Probability Distributions – Poisson Framework

5.1 Limiting Case of the Binomial Distribution

The Poisson distribution models the number of events occurring within a fixed interval of time or space. It can be derived as a limiting case of the Binomial distribution when the number of trials n grows very large, while the success probability p becomes very small, such that the product $n \cdot p = \lambda$ remains constant.

Theorem 5.1 (Poisson Limit Theorem):

As $n \rightarrow \infty$ and $p \rightarrow 0$ with $n \cdot p = \lambda$ remaining fixed, the Binomial PMF approaches the Poisson PMF:

$$P(X = x) = [e^{-\lambda} \cdot \lambda^x] / x! \quad \text{for } x = 0, 1, 2, \dots$$

Proof: Substitute $p = \lambda / n$ into the Binomial PMF formula:

$$P(X = x) = \{ n! / [x!(n-x)!] \} \cdot (\lambda / n)^x \cdot (1 - \lambda / n)^{n-x}$$

Rearranging the terms and expanding the factorial:

$$P(X = x) = [\lambda^x / x!] \cdot [n(n-1)(n-2)\dots(n-x+1) / n^x] \cdot (1 - \lambda / n)^n \cdot (1 - \lambda / n)^{-x}$$

Now, evaluate the limit of each term as $n \rightarrow \infty$:

1. The factorial fraction can be split into x individual linear terms:

$$\lim_{(n \rightarrow \infty)} [(n/n) \cdot ((n-1)/n) \cdot ((n-2)/n) \dots ((n-x+1)/n)] = 1 \cdot 1 \cdot 1 \dots 1 = 1$$

2. By the definition of the natural exponential function:

$$\lim_{(n \rightarrow \infty)} (1 - \lambda / n)^n = e^{-\lambda}$$

3. Since x is a fixed value:

$$\lim_{(n \rightarrow \infty)} (1 - \lambda / n)^{-x} = (1 - 0)^{-x} = 1$$

Combining these limits gives the Poisson PMF formula: $P(X = x) = [e^{-\lambda} \cdot \lambda^x] / x!$.

5.2 Step-by-Step Mathematical Proof of Moments

5.2.1 Expected Value (Mean) Derivation

Using the definition of expectation for a discrete random variable:

$$E[X] = \sum_{(x=0 \text{ to } \infty)} x \cdot [e^{-\lambda} \cdot \lambda^x] / x!$$

The term for $x = 0$ is 0, so we start the summation at $x = 1$. Since $x / x! = 1 / (x - 1)!$, we can rewrite the expression as:

$$E[X] = \sum_{(x=1 \text{ to } \infty)} [e^{-\lambda} \cdot \lambda^x] / (x - 1)!$$

Factor out λ and $e^{-\lambda}$ from the summation:

$$E[X] = \lambda \cdot e^{-\lambda} \cdot \sum_{(x=1 \text{ to } \infty)} \lambda^{x-1} / (x - 1)!$$

Let $y = x - 1$. As x ranges from 1 to ∞ , y ranges from 0 to ∞ :

$$E[X] = \lambda \cdot e^{-\lambda} \cdot \sum_{(y=0 \text{ to } \infty)} \lambda^y / y!$$

The summation is the Taylor series expansion for e^{λ} . Substituting this back gives:

$$E[X] = \lambda \cdot e^{-\lambda} \cdot e^{\lambda} = \lambda \cdot 1 = \lambda$$

$$\text{Mean: } \mu = E[X] = \lambda$$

5.2.2 Variance Derivation

Like before, we first evaluate the factorial expectation $E[X(X-1)]$:

$$E[X(X-1)] = \sum_{(x=0 \text{ to } \infty)} x(x-1) \cdot [e^{-\lambda} \cdot \lambda^x] / x!$$

The terms for $x = 0$ and $x = 1$ are 0, so we start the summation at $x = 2$. Since $x(x-1) / x! = 1 / (x-2)!$, we get:

$$E[X(X-1)] = \sum_{(x=2 \text{ to } \infty)} [e^{-\lambda} \cdot \lambda^x] / (x-2)!$$

Factor out λ^2 and $e^{-\lambda}$:

$$E[X(X-1)] = \lambda^2 \cdot e^{-\lambda} \cdot \sum_{(x=2 \text{ to } \infty)} \lambda^{x-2} / (x-2)!$$

Let $y = x - 2$. The summation simplifies to the Taylor series expansion for e^{λ} :

$$E[X(X-1)] = \lambda^2 \cdot e^{-\lambda} \cdot e^{\lambda} = \lambda^2$$

Now, calculate $E[X^2]$:

$$E[X^2] = E[X(X-1)] + E[X] = \lambda^2 + \lambda$$

Substitute this into the variance formula:

$$\text{Var}(X) = E[X^2] - (E[X])^2 = (\lambda^2 + \lambda) - \lambda^2 = \lambda$$

$$\text{Variance: } \sigma^2 = \lambda$$

The Poisson distribution has the unique structural property that its mean and variance are equal.

5.3 Extensive Problem Set: Poisson Distribution Problems

Problem 5.1

A network server receives web requests at an average rate of 4 requests per millisecond. Assuming the arrival of requests follows a Poisson process, calculate the probability that during a given millisecond, the server receives: 1. Exactly 2 requests. 2. No requests. 3. At least 3 requests.

Solution:

Let X be the number of requests per millisecond. Here, $X \sim \text{Poisson}(\lambda = 4)$.

1. Probability of exactly 2 requests ($P(X = 2)$):

$$P(X = 2) = [e^{-4} \cdot 4^2] / 2! = [0.0183156 \cdot 16] / 2 = 0.1465$$

2. Probability of no requests ($P(X = 0)$):

$$P(X = 0) = [e^{-4} \cdot 4^0] / 0! = 0.0183 \cdot 1 / 1 = 0.0183$$

3. Probability of at least 3 requests ($P(X \geq 3)$):

Using the complement rule:

$$P(X \geq 3) = 1 - P(X \leq 2) = 1 - [P(X = 0) + P(X = 1) + P(X = 2)]$$

Let's first calculate $P(X = 1)$:

$$P(X = 1) = [e^{-4} \cdot 4^1] / 1! = 0.0183156 \cdot 4 = 0.0733$$

Now substitute the values back into the equation:

$$P(X \geq 3) = 1 - [0.0183 + 0.0733 + 0.1465] = 1 - 0.2381 = 0.7619$$

Problem 5.2

Flaws occur randomly in a fiber-optic cable at an average rate of 0.6 flaws per kilometer. Find the probability of finding: 1. No flaws in a 2-kilometer section of cable. 2. Exactly 2 flaws in a 5-kilometer section of cable.

Solution:

The Poisson parameter scales linearly with the size of the interval:

1. For a 2-km section: The adjusted mean is $\lambda = 0.6 \cdot 2 = 1.2$ flaws.

$$P(X = 0) = [e^{-1.2} \cdot 1.2^0] / 0! = e^{-1.2} \approx 0.3012$$

2. For a 5-km section: The adjusted mean is $\lambda = 0.6 \cdot 5 = 3.0$ flaws.

$$P(X = 2) = [e^{-3} \cdot 3^2] / 2! = [0.049787 \cdot 9] / 2 \approx 0.2240$$

Problem 5.3

A warehouse tracking system shows that the probability of an automated scanner failing to read a barcode is **0.002**. If 1500 packages are scanned independently during a shift, use the Poisson approximation to find the probability that exactly 4 scanning failures occur.

Solution:

The number of scans $n = 1500$ is large, and the failure probability $p = 0.002$ is small. This fits the criteria for the Poisson approximation of a Binomial distribution. First, calculate the mean parameter λ :

$$\lambda = n \cdot p = 1500 \cdot 0.002 = 3.0$$

Now, evaluate the Poisson PMF at $x = 4$:

$$P(X = 4) = [e^{-3} \cdot 3^4] / 4! = [0.049787 \cdot 81] / 24 \approx 0.1680$$

Problem 5.4

An industrial fluid pump experiences unexpected operational shutdowns at an average rate of 2.5 shutdowns per year. Find the probability that the pump operates with at most 1 shutdown over a given 6-month period.

Solution:

The average rate is 2.5 shutdowns per year. Since a 6-month period is 0.5 years, we scale the parameter linearly:

$$\lambda = 2.5 \cdot 0.5 = 1.25$$

We want to find the probability of at most 1 shutdown, $P(X \leq 1) = P(X = 0) + P(X = 1)$:

$$P(X = 0) = [e^{-1.25} \cdot (1.25)^0] / 0! = e^{-1.25} \approx 0.2865$$

$$P(X = 1) = [e^{-1.25} \cdot (1.25)^1] / 1! = 0.2865 \cdot 1.25 \approx 0.3581$$

$$P(X \leq 1) = 0.2865 + 0.3581 = 0.6446$$

Problem 5.5

An IT service center logs data entry errors following a Poisson distribution. If the variance of the number of errors logged per day is exactly 4.0, calculate the probability that exactly 5 errors are logged on a given day.

Solution:

A key property of the Poisson distribution is that its mean and variance are equal ($\mu = \sigma^2 = \lambda$). Since we are given a variance of 4.0, the distribution parameter must be $\lambda = 4.0$.

Now, evaluate the Poisson PMF at $x = 5$:

$$P(X = 5) = [e^{-4} \cdot 4^5] / 5! = [0.0183156 \cdot 1024] / 120 \approx 0.1563$$

Chapter 6: Continuous Probability Distributions – Normal Framework

6.1 Mathematical Definition and Properties

Continuous random variables can take on any real value within a continuous range. Instead of a probability mass function, their behavior is defined by a **Probability Density Function (PDF)**, denoted $f(x)$, where the probability of an event is the area under the curve:

$$P(a \leq X \leq b) = \int_{(a \text{ to } b)} f(x) dx$$

Definition 6.1 (Normal Distribution): A continuous random variable X follows a Normal (or Gaussian) distribution with mean μ and variance σ^2 , written $X \sim N(\mu, \sigma^2)$, if its probability density function is given by:

$$f(x) = [1 / (\sigma \cdot \sqrt{2\pi})] \cdot e^{-(x - \mu)^2 / (2\sigma^2)} \quad \text{for } -\infty < x < \infty$$

Key Visual and Structural Characteristics:

- The distribution curve is symmetric and bell-shaped, centered directly around its mean μ .
- The mean, median, and mode of the distribution are identical.
- The total area under the density curve is exactly 1.0.
- **The Empirical Rule:** Approximately **68.27%** of the data falls within one standard deviation of the mean ($\mu \pm \sigma$), **95.45%** falls within two standard deviations ($\mu \pm 2\sigma$), and **99.73%** falls within three standard deviations ($\mu \pm 3\sigma$).

6.2 The Standard Normal Distribution Transformation

Because the normal PDF cannot be integrated analytically in closed form, probabilities are computed using a standardized transformation. This scales the general distribution $X \sim N(\mu, \sigma^2)$ into the **Standard Normal Distribution**, denoted by the variable Z , where the mean is 0 and the variance is 1 ($Z \sim N(0, 1)$).

Standardization Formula:

$$Z = (X - \mu) / \sigma$$

The cumulative density function (CDF) of the standard normal distribution is denoted by $\Phi(z) = P(Z \leq z)$. Values for $\Phi(z)$ are looked up in standard normal probability tables.

Using the symmetry of the distribution, we can derive the following useful properties:

- $\Phi(-z) = 1 - \Phi(z)$

$$\bullet P(a \leq X \leq b) = P((a-\mu)/\sigma \leq Z \leq (b-\mu)/\sigma) = \Phi((b-\mu)/\sigma) - \Phi((a-\mu)/\sigma)$$

6.3 Extensive Problem Set: Normal Distribution Problems

Standard Normal Cumulative Values for Reference:

$\Phi(1.00) = 0.8413$, $\Phi(1.50) = 0.9332$, $\Phi(2.00) = 0.9772$, $\Phi(2.50) = 0.9938$, $\Phi(0.50) = 0.6915$, $\Phi(1.25) = 0.8944$.

Problem 6.1

The tensile strength of a structural steel alloy is modeled by a normal distribution with a mean of 400 MPa and a standard deviation of 20 MPa. Find the probability that a randomly selected sample has a tensile strength: 1. Less than 370 MPa. 2. Between 380 MPa and 430 MPa.

Solution:

We are given $X \sim N(\mu = 400, \sigma^2 = 20^2)$, so $\sigma = 20$.

1. Probability strength is less than 370 MPa ($P(X < 370)$):

Standardize the value using the transformation formula:

$$Z = (370 - 400) / 20 = -30 / 20 = -1.50$$

$$P(X < 370) = P(Z < -1.50) = \Phi(-1.50)$$

Using the symmetry property:

$$\Phi(-1.50) = 1 - \Phi(1.50) = 1 - 0.9332 = 0.0668$$

2. Probability strength is between 380 MPa and 430 MPa ($P(380 \leq X \leq 430)$):

Standardize both boundary values:

$$Z_1 = (380 - 400) / 20 = -20 / 20 = -1.00$$

$$Z_2 = (430 - 400) / 20 = 30 / 20 = 1.50$$

Calculate the probability between these Z-scores:

$$P(-1.00 \leq Z \leq 1.50) = \Phi(1.50) - \Phi(-1.00) = \Phi(1.50) - [1 - \Phi(1.00)]$$

$$P(-1.00 \leq Z \leq 1.50) = 0.9332 - [1 - 0.8413] = 0.9332 - 0.1587 = 0.7745$$

Problem 6.2

An electronic component's operational life span follows a normal distribution. If the mean life span is 1200 hours and the standard deviation is 100 hours, what percentage of components will last longer than 1400 hours?

Solution:

We are given $X \sim N(\mu = 1200, \sigma = 100)$. We want to find $P(X > 1400)$.

Standardize the value:

$$Z = (1400 - 1200) / 100 = 200 / 100 = 2.00$$

Using the complement rule:

$$P(X > 1400) = P(Z > 2.00) = 1 - \Phi(2.00) = 1 - 0.9772 = 0.0228$$

Converting to a percentage, approximately **2.28%** of the components will last longer than 1400 hours.

Problem 6.3

An automated filling machine dispenses liquid soap into bottles. The volume dispensed follows a normal distribution with a mean of 500 mL. If the system must ensure that no more than **0.62%** of bottles contain less than 450 mL, what is the maximum allowable standard deviation σ ?

Solution:

We are given $X \sim N(\mu = 500, \sigma^2)$ and the condition $P(X < 450) \leq 0.0062$. Let's find the Z-score that corresponds to a cumulative probability of 0.0062:

$$P(Z < z) = 0.0062 \Rightarrow P(Z > -z) = 0.0062 \Rightarrow \Phi(-z) = 1 - 0.0062 = 0.9938$$

Looking up 0.9938 in our reference values gives a Z-score of 2.50. Because this is in the left tail, the target Z-score is **-2.50**.

Now, set up the standardization equation and solve for σ :

$$-2.50 = (450 - 500) / \sigma$$

$$-2.50 = -50 / \sigma \Rightarrow \sigma = -50 / -2.50 = 20$$

The maximum allowable standard deviation is **20 mL**.

Problem 6.4

The time it takes a technician to assemble a complex mechanical subsystem is normally distributed with a mean of 45 minutes and a variance of 16 minutes². Find the probability that a technician assembles a subsystem in less than 50 minutes.

Solution:

The variance is given as 16, so the standard deviation is $\sigma = \sqrt{16} = 4$ minutes. The mean is $\mu = 45$ minutes, giving $X \sim N(45, 4^2)$. We want to find $P(X < 50)$.

Standardize the boundary value:

$$Z = (50 - 45) / 4 = 5 / 4 = 1.25$$

Look up the cumulative probability for this Z-score:

$$P(X < 50) = \Phi(1.25) = 0.8944$$

There is an **89.44%** probability that the assembly takes less than 50 minutes.

Problem 6.5

A manufacturing group makes ball bearings whose diameters follow a normal distribution with a mean of 10.00 mm and a standard deviation of 0.02 mm. A bearing is considered within specification if its diameter is between 9.96 mm and 10.01 mm. Find the percentage of bearings that meet the specifications.

Solution:

We are given $X \sim N(\mu = 10.00, \sigma = 0.02)$. We want to find $P(9.96 \leq X \leq 10.01)$.

Standardize both boundary values:

$$Z_1 = (9.96 - 10.00) / 0.02 = -0.04 / 0.02 = -2.00$$

$$Z_2 = (10.01 - 10.00) / 0.02 = 0.01 / 0.02 = 0.50$$

Calculate the probability between these Z-scores:

$$P(-2.00 \leq Z \leq 0.50) = \Phi(0.50) - \Phi(-2.00) = \Phi(0.50) - [1 - \Phi(2.00)]$$

$$P(-2.00 \leq Z \leq 0.50) = 0.6915 - [1 - 0.9772] = 0.6915 - 0.0228 = 0.6687$$

Converting to a percentage, **66.87%** of the manufactured bearings meet the specifications.

Summary of Distribution Models and Parameters

To help choose the right model for a given problem, the following table summarizes the structural formulas, parameters, and moments for the three theoretical distributions covered in these notes:

Distribution Model	Type	Probability Function (PMF / PDF)	Parameters	Mean (μ)	Variance (σ^2)
Binomial	Discrete	${}_nC_x \cdot p^x \cdot q^{n-x}$	n (trials), p (success prob)	$n \cdot p$	$n \cdot p \cdot q$
Poisson	Discrete	$[e^{-\lambda} \cdot \lambda^x] / x!$	λ (average arrival rate)	λ	λ
Normal	Continuous	$[1 / (\sigma\sqrt{2\pi})] \cdot e^{-(x-\mu)^2 / (2\sigma^2)}$	μ (mean), σ (std dev)	μ	σ^2